CUNEF UNIVERSIDAD

Econometrics 2024/2025

Multiple regression: additional aspects

Outline

More on Functional Form

More on goodness-of-fit and selection of regressors

Prediction

Outline

More on Functional Form

More on goodness-of-fit and selection of regressors

Prediction

Multiple Regression Analysis: Further Issues

More on Functional Form:

- More on using logarithmic functional forms:
- Convenient percentage/elasticity interpretation
- Slope coefficients of logged variables are invariant to rescalings
- Taking logs often eliminates/mitigates problems with outliers
- Taking logs often helps to secure normality and homoskedasticity
- Variables measured in units such as years should not be logged
- Variables measured in percentage points should also not be logged
- Logs must not be used if variables take on zero or negative values
- It is hard to reverse the log-operation when constructing predictions

Using quadratic functional forms

- Example: Wage equation:

Concave experience profile

$$\widehat{wage} = \underset{(.35)}{3.73} + \underset{(.041)}{.298}\, exper - \underset{(.0009)}{.0061}\, exper^2$$

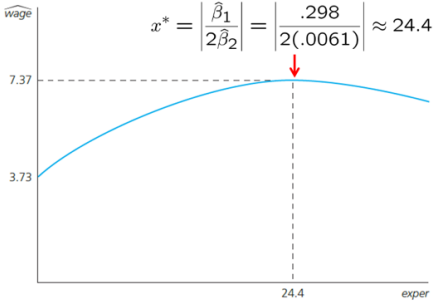
$$n = 526, R^2 = .093$$

- Marginal effect of experience

$$\frac{\Delta wage}{\Delta exper} = .298 - 2(.0061)exper$$

The first year of experience increases the wage by some \$.30, the second year by .298 - 2(.0061)(1) = \$.29 etc.

Wage maximum with respect to work experience



- Does this mean the return to experience becomes negative after 24.4 years?
- Not necessarily. It depends on how many observations in the sample lie to the right of the turnaround point.
- In the given example, these are about 28% of the observations. There may be a specification problem (e.g. omitted variables).

Example: Effects of pollution on housing prices

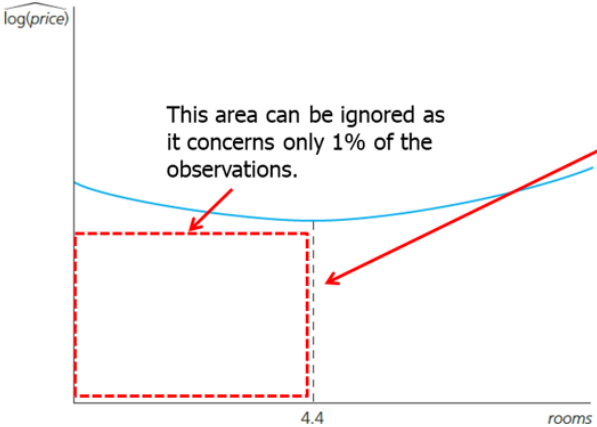
$$\log(\widehat{price}) = \underset{(.57)}{13.39} - \underset{(.115)}{.902}\log(nox) - \underset{(.043)}{.087}\log(dist)$$
$$-\underset{(.165)}{.545}\,rooms + \underset{(.013)}{.062}\,rooms^2 - \underset{(.006)}{048}\,stratio$$
$$n = 506,\ R^2 = .603$$

nox: nitrogen oxide in the air
$dist$: distance from employment centers
$rooms$: number of rooms
$stratio$: average student/teacher ratio

Does this mean that, at a low number of rooms, more rooms are associated with lower prices?

$$\Rightarrow \quad \frac{\Delta\log(price)}{\Delta rooms} = \frac{\%\Delta price}{\Delta rooms} = -.545 + .124 rooms$$

Calculation of the turnaround point



Turnaround point:

$$x^* = \left| \frac{-.545}{2(.062)} \right| \approx 4.4$$

Increase rooms from 5 to 6:

$$-.545 + .124(5) = +7.5\% \ price$$

Increase rooms from 6 to 7:

$$-.545 + .124(6) = +19.9\% \ price$$

Other possibilities

- Another model:

$$\log(price) = \beta_0 + \beta_1 \log(nox) + \beta_2 [\log(nox)]^2 + \beta_3 crime + \beta_4 rooms + \beta_5 rooms^2 + \beta_6 stratio + u$$

$$\Rightarrow \quad \frac{\Delta \log(price)}{\Delta \log(nox)} = \frac{\%\Delta price}{\%\Delta nox} = \beta_1 + 2\beta_2 [\log(nox)]$$

- Highter polinomials:

$$cost = \beta_0 + \beta_1 quantity + \beta_2 quantity^2 + \beta_3 quantity^3 + u$$

Models with interaction terms

- Interaction term: sqrft *bdrms

$$price = \beta_0 + \beta_1 sqrft + \beta_2 bdrms + \beta_3 sqrft * bdrms + \beta_4 bthrms + u$$

$$\Rightarrow \frac{\Delta price}{\Delta bdrms} = \beta_2 + \beta_3 sqrft$$

- Interaction effects complicate interpretation of parameters
  $\beta_2 =$ Effect of bedrooms, but not for a square footage of zero

Reparametrization of interaction effects

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u$$

Population means; may be replaced by sample means

$$y = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + u$$

Effect of x_2 if all variables take on their mean values

Advantages of reparametrization:

- Easy interpretation of all parameters
- Standard errors for partial effects at the mean values available
- If necessary, interaction may be centered at other interesting values

Average partial effects

- In models with quadratics, interactions, and other nonlinear functional forms, the partial effect depend on the values of one or more explanatory variables
- Average partial effect (APE) is a summary measure to describe the relationship between dependent variable and each explanatory variable
- After computing the partial effect and plugging in the estimated parameters, average the partial effects for each unit across the sample

Outline

More on Functional Form

More on goodness-of-fit and selection of regressors

Prediction

More on goodness-of-fit and selection of regressors

General remarks on R-squared

- A high R-squared does not imply that there is a causal interpretation
- A low R-squared does not preclude precise estimation of partial effects
- Adjusted R-squared
- What is the ordinary R-squared supposed to measure?
- $R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{SSR/n}{SST/n}$ is an estimator for $1 - \frac{\sigma_u^2}{\sigma_y^2}$ (population R-squared)

Adjusted R-squared

- A better estimate taking into account degrees of freedom would be

$$\bar{R}^2 = 1 - \frac{(SSR/(n-k-1))}{(SST/(n-1))} = adjusted\ R^2$$

  - The adjusted R-squared imposes a penalty for adding new regressors
  - The adjusted R-squared increases if, and only if, the t-statistic of a newly added regressor is greater than one in absolute value
- Relationship between R-squared and adjusted R-squared

$$\bar{R}^2 = 1 - (1 - R^2)(n-1)/(n-k-1)$$

(The adjusted R-squared may even get negative!)

Using adjusted R-squared to choose between nonnested models

- Models are nonnested if neither model is a special case of the other

$$rdintens = \beta_0 + \beta_1 \log(sales) + u$$

$\rightarrow R^2 = .061,\ \bar{R}^2 = .030$

$$rdintens = \beta_0 + \beta_1 sales + \beta_2 sales^2 + u$$

$\rightarrow R^2 = .148,\ \bar{R}^2 = .090$

- A comparison between the R-squared of both models would be unfair to the first model because the first model contains fewer parameters
- In the given example, even after adjusting for the difference in degrees of freedom, the quadratic model is preferred

Comparing models with different dependent variables

- R-squared or adjusted R-squared must not be used to compare models which differ in their definition of the dependent variable
- Example: CEO compensation and firm performance

There is much less variation in log(salary) that needs to be explained than in salary

$$\widehat{salary} = \underset{(223.90)}{830.63} + \underset{(.0089)}{.0163}\, sales + \underset{(11.08)}{19.03}\, roe$$

$$n = 209, R^2 = .029, \bar{R}^2 = .020, SST = 391{,}732{,}982$$

$$\widehat{lsalary} = \underset{(0.29)}{4.36} + \underset{(.033)}{.275}\, sales + \underset{(.0040)}{.0179}\, roe$$

$$n = 209, R^2 = .282, \bar{R}^2 = .275, SST = 66.72$$

Controlling for too many factors in regression analysis

- In some cases, certain variables should not be held fixed
  - In a regression of traffic fatalities on state beer taxes (and other factors) one should not directly control for beer consumption
  - In a regression of family health expenditures on pesticide usage among farmers one should not control for doctor visits
- Different regressions may serve different purposes
  - In a regression of house prices on house characteristics, one would only include price assessments if the purpose of the regression is to study their validity; otherwise one would not include them

Adding regressors to reduce the error variance

- Adding regressors may excarcerbate multicollinearity problems
- On the other hand, adding regressors reduces the error variance
- Variables that are uncorrelated with other regressors should be added because they reduce error variance without increasing multicollinearity
- However, such uncorrelated variables may be hard to find
- Example: Individual beer consumption and beer prices
- Including individual characteristics in a regression of beer consumption on beer prices leads to more precise estimates of the price elasticity

Outline

More on Functional Form

More on goodness-of-fit and selection of regressors

Prediction

Predicting when the dependent variable is logarithmic

- Predicting y when $\log(y)$ is the dependent variable

$$\log(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u$$

$$\Rightarrow y = \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k)\exp(u)$$

- Under the additional assumption that u is independent of $x_1, \ldots, x_k$:

$$\Rightarrow \mathrm{E}(y|\mathbf{x}) = \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k)\mathrm{E}(\exp(u))$$

$$\Rightarrow \hat{y} = \exp(\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_k x_k)(\frac{1}{n}\sum_{i=1}^{n}\exp(\hat{u}_i))$$

CUNEF
UNIVERSIDAD

Campus Almansa | Calle Almansa 101 | 28040 Madrid

cunef.edu